

38 — MAXIMUM LIKELIHOOD ESTIMATION

Aim

To estimate the mean and variance of a Gaussian distribution using Maximum Likelihood Estimation and compare the MLE mean with the least-squares estimate.

Software Required

- Python 3.x
- Jupyter Notebook / Google Colab
- NumPy

Algorithm + Procedure

1. Generate random samples from a Gaussian distribution.
2. Calculate the sample mean.
3. Calculate the sample variance.
4. Treat the sample mean as the MLE of the Gaussian mean.
5. Generate candidate mean values.
6. Calculate the SSE for each candidate.
7. Find the candidate having minimum SSE.
8. Compare it with the MLE mean.
9. Display the results.

Code

```
import numpy as np
np.random.seed(1)
data = np.random.normal(
    loc=50,
    scale=5,
    size=200
)
mle_mean = np.mean(data)
mle_variance = np.var(data)
print(f"MLE Mean = {mle_mean:.3f}")
```

```

print(f'MLE Variance = {mle_variance:.3f}')
candidates = np.linspace(45, 55, 100)
sse = [
    np.sum((data - c) ** 2)
    for c in candidates
]
best_candidate = candidates[np.argmin(sse)]
print(
    f'Candidate minimising SSE: "
    f"{best_candidate:.3f}"
)

```

Output

```

MLE Mean = 50.533
MLE Variance = 20.705
Candidate minimising SSE: 50.556

```

Result

The Maximum Likelihood Estimate was successfully calculated. The MLE mean was found to be close to the value that minimised the sum of squared errors.